

Heat Transfer Operation

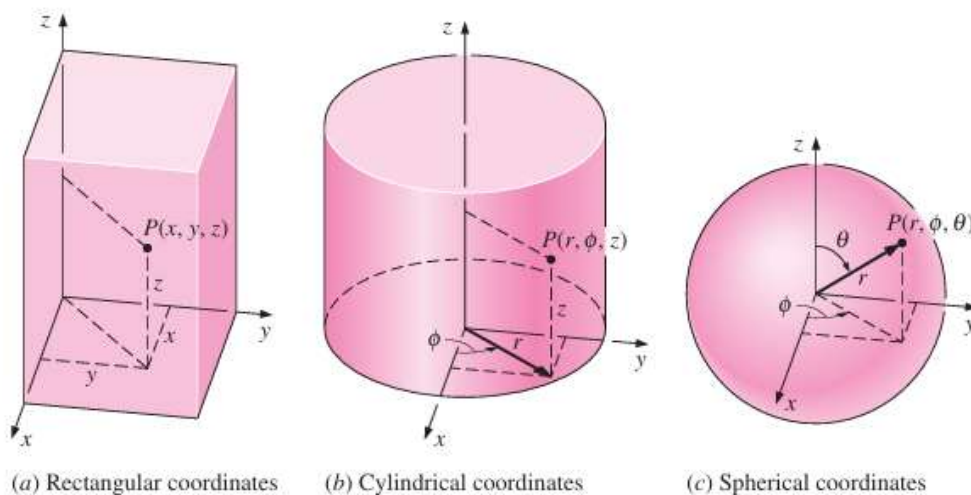
Unit 2. Conduction

- Mechanism of heat conduction
- Fourier's law, thermal conductivity of solids, liquids and gases, effect of temperature on thermal conductivity
- Steady state Heat conduction
- Development of equations for conduction through plane and curved surfaces like spheres, cylinders etc.
- Compound resistances
- Critical thickness of insulation
- Heat transfer with internal heat generation
- Introduction to transient heat conduction
- Introduction to Multidimensional heat conduction.

- Heat conduction is defined as the transfer of thermal energy from the more energetic particles of a medium to the adjacent less energetic ones.
- Conduction can take place in liquids and gases as well as solids provided that there is no bulk motion involved.
- Although heat transfer and temperature are closely related, they are of a different nature. Unlike temperature, heat transfer has direction as well as magnitude, and **thus heat transfer is a vector quantity.**

Notations

- The location of a point is specified as (x, y, z) in rectangular coordinates, as (r, ϕ, z) in cylindrical coordinates, and as (r, ϕ, θ) in spherical coordinates, where the distances x, y, z , and r and the angles ϕ and θ are as shown in Figure.
- Then the temperature at a point (x, y, z) at time t in rectangular coordinates is expressed as $T(x, y, z, t)$.
- The best coordinate system for a given geometry is the one that describes the surfaces of the geometry best.



Notes created by Dr. Panchami Patel

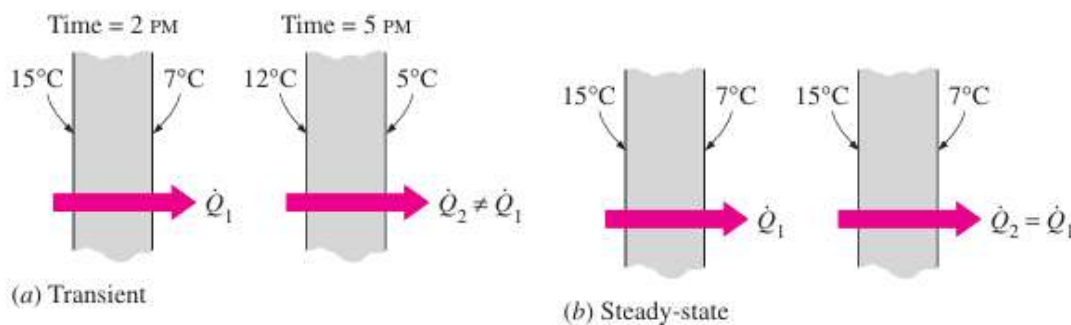
Reference Books:

- Cengel A. Yunus. "Heat Transfer – A Practical Approach", McGraw Hill.

Heat Transfer Operation

Steady versus Transient Heat Transfer

- Heat transfer problems are often classified as being steady (also called steady state) or transient (also called unsteady).
- The term steady implies no change with time at any point within the medium, while transient implies variation with time or time dependence.
- Therefore, the temperature or heat flux remains unchanged with time during steady heat transfer through a medium at any location, although both quantities may vary from one location to another.
- Heat transfer through the walls of a house will be steady when the conditions inside the house and the outdoors remain constant for several hours. But even in this case, the temperatures on the inner and outer surfaces of the wall will be different unless the temperatures inside and outside the house are the same.
- The cooling of an apple in a refrigerator, on the other hand, is a transient heat transfer process since the temperature at any fixed point within the apple will change with time during cooling.
- Most heat transfer problems encountered in practice are transient in nature, but they are usually analyzed under some presumed steady conditions since steady processes are easier to analyze.



Multidimensional Heat Transfer

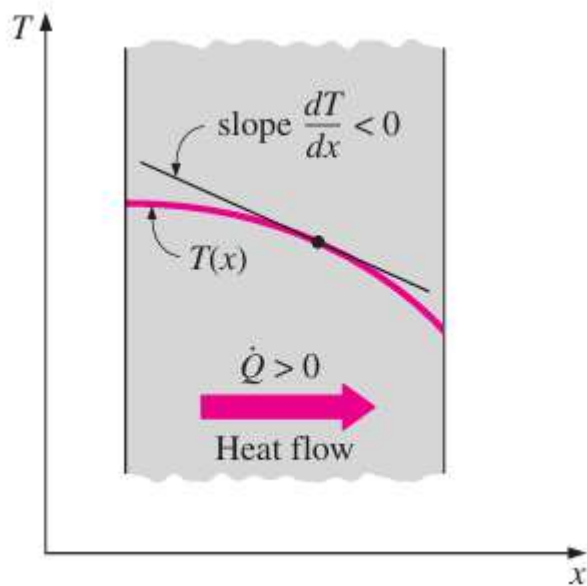
- Heat transfer problems are also classified as being one-dimensional, two dimensional, or three-dimensional, depending on the relative magnitudes of heat transfer rates in different directions.
- In the most general case, heat transfer through a medium is three-dimensional.
- That is, the temperature varies along all three primary directions within the medium during the heat transfer process.

Differential form of Fourier's law of heat conduction for one-dimensional heat conduction:

$$\dot{Q}_{\text{cond}} = -kA \frac{dT}{dx}$$

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where k is the thermal conductivity of the material, which is a measure of the ability of a material to conduct heat, and dT/dx is the temperature gradient, which is the slope of the temperature curve on a T - x diagram.



Heat is conducted in the direction of decreasing temperature, and thus the temperature gradient is negative when heat is conducted in the positive x -direction. The negative sign in Eq. ensures that heat transfer in the positive x -direction is a positive quantity.

Isotropic vs. anisotropic:

- Most engineering materials are isotropic in nature, and thus they have the same properties in all directions.
- For such materials we do not need to be concerned about the variation of properties with direction.
- But in anisotropic materials such as the fibrous or composite materials, the properties may change with direction.
- For example, some of the properties of wood along the grain are different than those in the direction normal to the grain.
- In such cases the thermal conductivity may need to be expressed as a tensor quantity to account for the variation with direction.

Heat Generation: A medium through which heat is conducted may involve the conversion of electrical, nuclear, or chemical energy into heat (or thermal) energy. In heat conduction analysis, such conversion processes are characterized as heat generation.

When the variation of heat generation with position is known, the total rate of heat generation in a medium of volume V can be determined from

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$$\dot{G} = \int_V \dot{g} dV$$

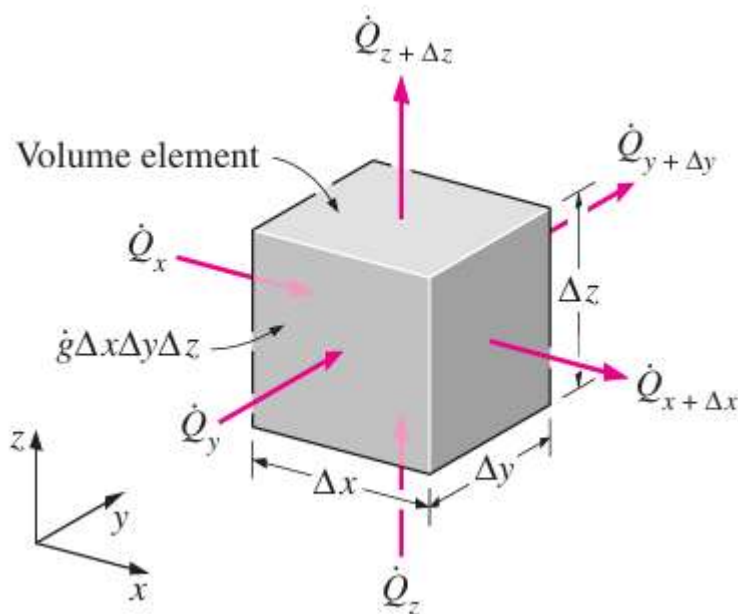
In the special case of uniform heat generation, $\dot{G} = \dot{g}V$ where $\dot{g}$ is the constant rate of heat generation per unit volume.

GENERAL HEAT CONDUCTION EQUATION

Multidimensional heat transfer

Rectangular Coordinates

Consider a small rectangular element of length x , width y , and height z . Assume the density of the body is ρ and the specific heat is C .



An energy balance on this element during a small time interval t can be expressed as

$$\left(\text{Rate of heat conduction at } x, y, \text{ and } z \right) - \left(\text{Rate of heat conduction at } x + \Delta x, y + \Delta y, \text{ and } z + \Delta z \right) + \left(\text{Rate of heat generation inside the element} \right) = \left(\text{Rate of change of the energy content of the element} \right)$$

$$\dot{Q}_x + \dot{Q}_y + \dot{Q}_z - \dot{Q}_{x+\Delta x} - \dot{Q}_{y+\Delta y} - \dot{Q}_{z+\Delta z} + \dot{G}_{\text{element}} = \frac{\Delta E_{\text{element}}}{\Delta t}$$

$$V_{\text{element}} = \Delta x \Delta y \Delta z$$

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Heat Transfer Operation

Change in the energy content of the element:

$$\Delta E_{\text{element}} = E_{t+\Delta t} - E_t = mC(T_{t+\Delta t} - T_t) = \rho C \Delta x \Delta y \Delta z (T_{t+\Delta t} - T_t)$$

Rate of heat generation within the element:

$$\dot{G}_{\text{element}} = \dot{g} V_{\text{element}} = \dot{g} \Delta x \Delta y \Delta z$$

Substituting in equation

$$\dot{Q}_x + \dot{Q}_y + \dot{Q}_z - \dot{Q}_{x+\Delta x} - \dot{Q}_{y+\Delta y} - \dot{Q}_{z+\Delta z} + \dot{g} \Delta x \Delta y \Delta z = \rho C \Delta x \Delta y \Delta z \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Dividing by $\Delta x \Delta y \Delta z$

$$-\frac{1}{\Delta y \Delta z} \frac{\dot{Q}_{x+\Delta x} - \dot{Q}_x}{\Delta x} - \frac{1}{\Delta x \Delta z} \frac{\dot{Q}_{y+\Delta y} - \dot{Q}_y}{\Delta y} - \frac{1}{\Delta x \Delta y} \frac{\dot{Q}_{z+\Delta z} - \dot{Q}_z}{\Delta z} + \dot{g} = \rho C \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Noting that the heat transfer areas of the element for heat conduction in the x, y, and z directions are $A_x = \Delta y \Delta z$, $A_y = \Delta x \Delta z$, and $A_z = \Delta x \Delta y$

From the definition of the derivative and Fourier's law of heat conduction

$$\begin{aligned} \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta y \Delta z} \frac{\dot{Q}_{x+\Delta x} - \dot{Q}_x}{\Delta x} &= \frac{1}{\Delta y \Delta z} \frac{\partial \dot{Q}_x}{\partial x} = \frac{1}{\Delta y \Delta z} \frac{\partial}{\partial x} \left(-k \Delta y \Delta z \frac{\partial T}{\partial x} \right) = -\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) \\ \lim_{\Delta y \rightarrow 0} \frac{1}{\Delta x \Delta z} \frac{\dot{Q}_{y+\Delta y} - \dot{Q}_y}{\Delta y} &= \frac{1}{\Delta x \Delta z} \frac{\partial \dot{Q}_y}{\partial y} = \frac{1}{\Delta x \Delta z} \frac{\partial}{\partial y} \left(-k \Delta x \Delta z \frac{\partial T}{\partial y} \right) = -\frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) \\ \lim_{\Delta z \rightarrow 0} \frac{1}{\Delta x \Delta y} \frac{\dot{Q}_{z+\Delta z} - \dot{Q}_z}{\Delta z} &= \frac{1}{\Delta x \Delta y} \frac{\partial \dot{Q}_z}{\partial z} = \frac{1}{\Delta x \Delta y} \frac{\partial}{\partial z} \left(-k \Delta x \Delta y \frac{\partial T}{\partial z} \right) = -\frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) \end{aligned}$$

Taking the limit as Δx , Δy , Δz and $\Delta t \rightarrow 0$

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{g} = \rho C \frac{\partial T}{\partial t}$$

is the general heat conduction equation in rectangular coordinates.

In the case of constant thermal conductivity

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{g}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

is known as the Fourier-Biot equation.

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where the property $k/\rho C$ is the thermal diffusivity of the material.

Equation reduced to

1. Steady-state: (called the Poisson equation)

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{g}}{k} = 0$$

2. Transient, no heat generation: (called the diffusion equation)

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

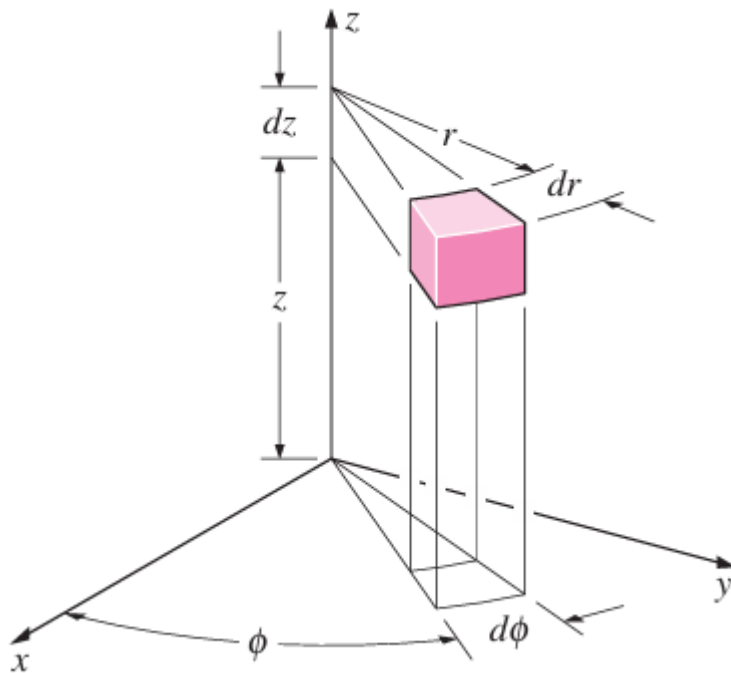
3. Steady-state, no heat generation: (called the Laplace equation)

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0$$

Cylindrical Coordinates

Relations between the coordinates of a point in rectangular and cylindrical coordinate systems:

$$x = r \cos \phi, \quad y = r \sin \phi, \quad \text{and} \quad z = z$$



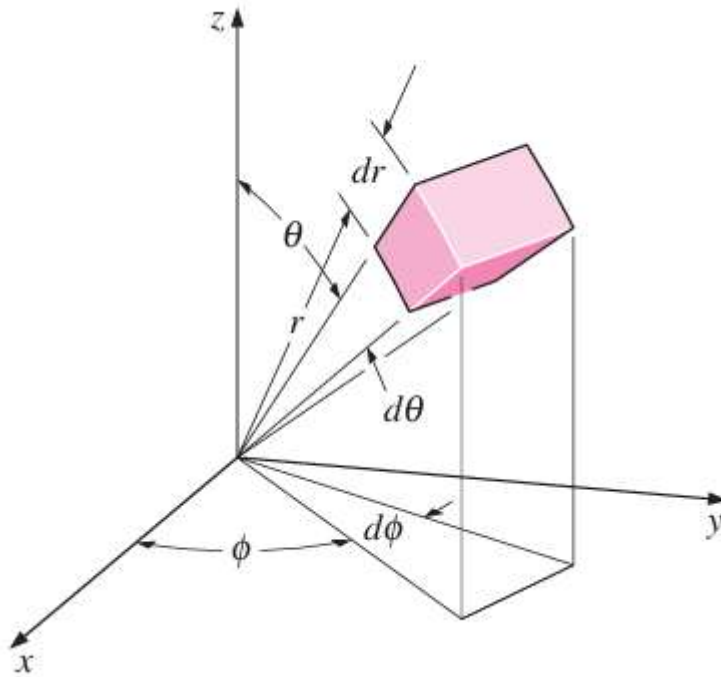
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$$\frac{1}{r} \frac{\partial}{\partial r} \left(kr \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \phi} \left(kr \frac{\partial T}{\partial \phi} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{g} = \rho C \frac{\partial T}{\partial t}$$

Spherical Coordinates

Relations between the coordinates of a point in rectangular and spherical coordinate systems:

$$x = r \cos \phi \sin \theta, \quad y = r \sin \phi \sin \theta, \quad \text{and} \quad z = r \cos \theta$$



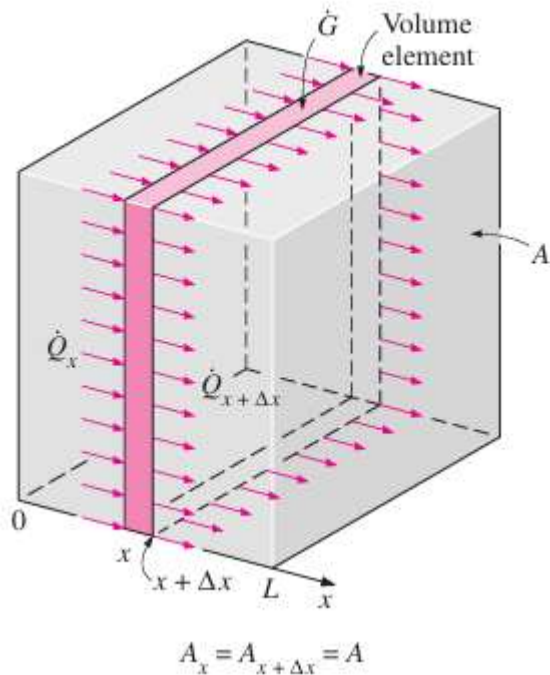
$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(kr^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial}{\partial \phi} \left(k \frac{\partial T}{\partial \phi} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(k \sin \theta \frac{\partial T}{\partial \theta} \right) + \dot{g} = \rho C \frac{\partial T}{\partial t}$$

Heat Transfer Operation

Heat Conduction Equation in a Large Plane Wall

Consider a thin element of thickness Δx in a large plane wall

Assume the density of the wall is ρ , the specific heat is C , and the area of the wall normal to the direction of heat transfer is A .



An energy balance on this thin element during a small time interval t can be expressed as

$$\left(\begin{array}{c} \text{Rate of heat} \\ \text{conduction} \\ \text{at } x \end{array} \right) - \left(\begin{array}{c} \text{Rate of heat} \\ \text{conduction} \\ \text{at } x + \Delta x \end{array} \right) + \left(\begin{array}{c} \text{Rate of heat} \\ \text{generation} \\ \text{inside the} \\ \text{element} \end{array} \right) = \left(\begin{array}{c} \text{Rate of change} \\ \text{of the energy} \\ \text{content of the} \\ \text{element} \end{array} \right)$$

$$\dot{Q}_x - \dot{Q}_{x+\Delta x} + \dot{G}_{\text{element}} = \frac{\Delta E_{\text{element}}}{\Delta t}$$

Change in the energy content of the element:

$$\Delta E_{\text{element}} = E_{t+\Delta t} - E_t = mC(T_{t+\Delta t} - T_t) = \rho CA\Delta x(T_{t+\Delta t} - T_t)$$

Rate of heat generation within the element:

$$\dot{G}_{\text{element}} = \dot{g}V_{\text{element}} = \dot{g}A\Delta x$$

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Substituting into Equation

$$\dot{Q}_x - \dot{Q}_{x+\Delta x} + \dot{g}A\Delta x = \rho CA\Delta x \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Dividing by $A\Delta x$ gives

$$-\frac{1}{A} \frac{\dot{Q}_{x+\Delta x} - \dot{Q}_x}{\Delta x} + \dot{g} = \rho C \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Taking the limit as $\Delta x \rightarrow 0$ and $\Delta t \rightarrow 0$ yields

$$\frac{1}{A} \frac{\partial}{\partial x} \left(kA \frac{\partial T}{\partial x} \right) + \dot{g} = \rho C \frac{\partial T}{\partial t}$$

Assuming constant thermal conductivity

$$\frac{\partial^2 T}{\partial x^2} + \frac{\dot{g}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

where the property $k/\rho C$ is the thermal diffusivity of the material and represents how fast heat propagates through a material.

Equation can reduce to the following forms:

1. Steady-state ($\frac{\partial}{\partial t} = 0$): $\frac{d^2 T}{dx^2} + \frac{\dot{g}}{k} = 0$
2. Transient, no heat generation ($\dot{g} = 0$): $\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$
3. Steady-state, no heat generation: ($\frac{\partial}{\partial t} = 0$ and $\dot{g} = 0$): $\frac{d^2 T}{dx^2} = 0$

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$$\frac{d^2T}{dx^2} = 0$$

On integration

$$\frac{dT}{dx} = C_1$$

On integration

$$T(x) = C_1x + C_2$$

Boundary condition 1: $x = 0$, $T = T_1$

$$\begin{array}{ccc} T(x) = C_1x + C_2 \\ \uparrow & & \uparrow \\ 0 & & 0 \\ \underbrace{} & & \\ T_1 & & \end{array}$$

$$T_1 = C_1 \times 0 + C_2 \rightarrow C_2 = T_1$$

Boundary condition 2: $x = L$, $T = T_2$

$$T(L) = C_1L + C_2 \rightarrow T_2 = C_1L + T_1 \rightarrow C_1 = \frac{T_2 - T_1}{L}$$

Substituting C_1 and C_2 in main equation

$$T(x) = \frac{T_2 - T_1}{L}x + T_1$$

Substituting T in Fourier's equation:

$$\dot{Q}_{\text{cond}} = -kA \frac{dT}{dx} = -kA \left(d \left(\left(\frac{T_2 - T_1}{L} \right) x + T_1 \right) \right) = k \times A \times \frac{T_1 - T_2}{L}$$

$$\dot{Q}_{\text{wall}} = -kA \frac{dT}{dx} = -kAC_1 = -kA \frac{T_2 - T_1}{L} = kA \frac{T_1 - T_2}{L}$$

Heat Transfer Operation

Heat Conduction Equation in a Long Cylinder

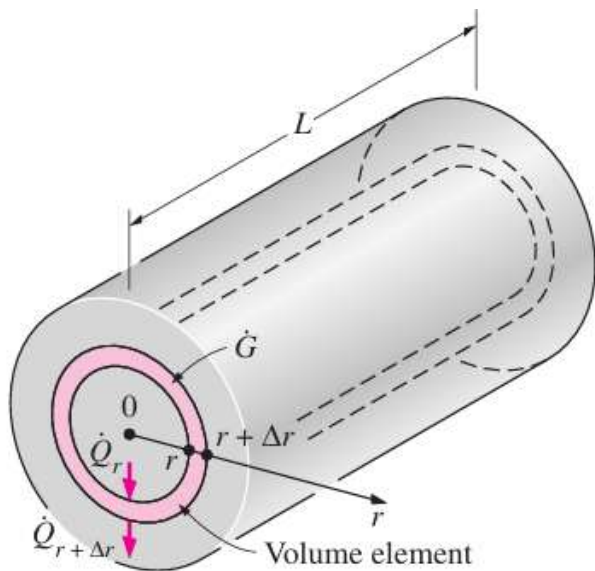
Consider a thin cylindrical shell element of thickness Δr in a long cylinder.

Assume the density of the cylinder is ρ , the specific heat is C , and the length is L .

The area of the cylinder normal to the direction of heat transfer at any location is $A=2\pi rL$, where r is the value of the radius at that location.

An energy balance on this thin cylindrical shell element during a small time interval Δt can be expressed as

$$\left(\text{Rate of heat conduction at } r \right) - \left(\text{Rate of heat conduction at } r + \Delta r \right) + \left(\text{Rate of heat generation inside the element} \right) = \left(\text{Rate of change of the energy content of the element} \right)$$



$$\dot{Q}_r - \dot{Q}_{r+\Delta r} + \dot{G}_{\text{element}} = \frac{\Delta E_{\text{element}}}{\Delta t}$$

$$\Delta E_{\text{element}} = E_{t+\Delta t} - E_t = mC(T_{t+\Delta t} - T_t) = \rho CA\Delta r(T_{t+\Delta t} - T_t)$$

$$\dot{G}_{\text{element}} = \dot{g}V_{\text{element}} = \dot{g}A\Delta r$$

Substituting

$$\dot{Q}_r - \dot{Q}_{r+\Delta r} + \dot{g}A\Delta r = \rho CA\Delta r \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

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Dividing the equation above by $A\Delta r$

$$-\frac{1}{A} \frac{\dot{Q}_{r+\Delta r} - \dot{Q}_r}{\Delta r} + \dot{g} = \rho C \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

From the definition of the derivative and Fourier's law of heat conduction

$$\lim_{\Delta r \rightarrow 0} \frac{\dot{Q}_{r+\Delta r} - \dot{Q}_r}{\Delta r} = \frac{\partial \dot{Q}}{\partial r} = \frac{\partial}{\partial r} \left(-kA \frac{\partial T}{\partial r} \right)$$

Taking the limit as $\Delta r \rightarrow 0$ and $\Delta t \rightarrow 0$

$$\frac{1}{A} \frac{\partial}{\partial r} \left(kA \frac{\partial T}{\partial r} \right) + \dot{g} = \rho C \frac{\partial T}{\partial t}$$

Noting that the heat transfer area in this case is $A = 2\pi rL$, the one dimensional transient heat conduction equation in a cylinder becomes

$$\frac{1}{r} \frac{\partial}{\partial r} \left(rk \frac{\partial T}{\partial r} \right) + \dot{g} = \rho C \frac{\partial T}{\partial t}$$

For the case of constant thermal conductivity

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{\dot{g}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

(1) *Steady-state:*
($\partial/\partial t = 0$)

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) + \frac{\dot{g}}{k} = 0$$

(2) *Transient, no heat generation:*
($\dot{g} = 0$)

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

(3) *Steady-state, no heat generation:*
($\partial/\partial t = 0$ and $\dot{g} = 0$)

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0$$

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0$$

Integrating the equation with respect to r

$$r \frac{dT}{dr} = C_1$$

Heat Transfer Operation

$$\frac{dT}{dr} = \frac{C_1}{r}$$

Integrating again with respect to r

$$T(r) = C_1 \ln r + C_2$$

apply both boundary conditions

$$T(r_1) = T_1 \rightarrow C_1 \ln r_1 + C_2 = T_1$$

$$T(r_2) = T_2 \rightarrow C_1 \ln r_2 + C_2 = T_2$$

Solving

$$C_1 = \frac{T_2 - T_1}{\ln(r_2/r_1)}$$

$$C_2 = T_1 - \frac{T_2 - T_1}{\ln(r_2/r_1)} \ln r_1$$

$$T(r) = \left(\frac{\ln(r/r_1)}{\ln(r_2/r_1)} \right) (T_2 - T_1) + T_1$$

Rate of heat transfer is determined from Fourier's law

$$\dot{Q}_{\text{cylinder}} = -kA \frac{dT}{dr} = -k(2\pi rL) \frac{C_1}{r} = -2\pi kLC_1 = 2\pi kL \frac{T_1 - T_2}{\ln(r_2/r_1)}$$

Heat Transfer Operation

Heat Conduction Equation in a Sphere

Consider a sphere with density ρ , specific heat C , and outer radius R .

The area of the sphere normal to the direction of heat transfer at any location is $A = 4\pi r^2$, where r is the value of the radius at that location.

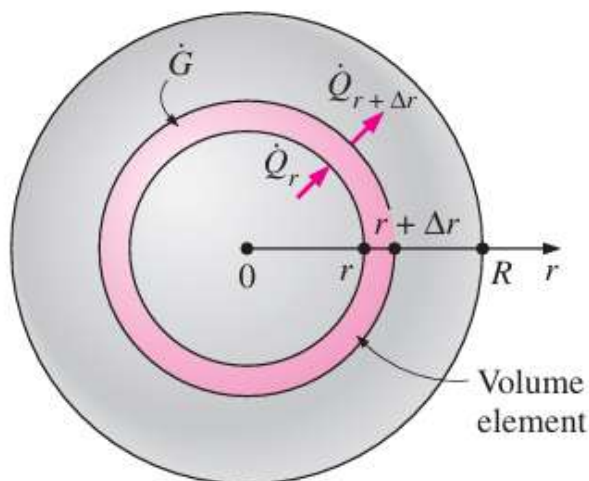
By considering a thin spherical shell element of thickness Δr and repeating the approach described above for the cylinder by using $A = 4\pi r^2$ instead of $A = 2\pi rL$.

The one-dimensional transient heat conduction equation for a sphere is

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 k \frac{\partial T}{\partial r} \right) + \dot{g} = \rho C \frac{\partial T}{\partial t}$$

In the case of constant thermal conductivity

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{\dot{g}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$



(1) *Steady-state:* $\left(\partial/\partial t = 0 \right)$ $\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) + \frac{\dot{g}}{k} = 0$

(2) *Transient, no heat generation:* $\left(\dot{g} = 0 \right)$ $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) = \frac{1}{\alpha} \frac{\partial T}{\partial t}$

(3) *Steady-state, no heat generation:* $\left(\partial/\partial t = 0 \text{ and } \dot{g} = 0 \right)$ $\frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0$ or $r \frac{d^2 T}{dr^2} + 2 \frac{dT}{dr} = 0$

Heat Transfer Operation

$$\frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0$$

Integrating the differential equation once with respect to r

$$r^2 \frac{dT}{dr} = C_1$$

divide both sides of this equation by r^2 to bring it to a readily integrable form

$$\frac{dT}{dr} = \frac{C_1}{r^2}$$

Again integrating with respect to r

$$T(r) = -\frac{C_1}{r} + C_2$$

apply both boundary conditions

$$T(r_1) = T_1 \rightarrow -\frac{C_1}{r_1} + C_2 = T_1$$

$$T(r_2) = T_2 \rightarrow -\frac{C_1}{r_2} + C_2 = T_2$$

$$C_1 = -\frac{r_1 r_2}{r_2 - r_1} (T_1 - T_2) \quad \text{and} \quad C_2 = \frac{r_2 T_2 - r_1 T_1}{r_2 - r_1}$$

$$T(r) = \frac{r_1 r_2}{r(r_2 - r_1)} (T_1 - T_2) + \frac{r_2 T_2 - r_1 T_1}{r_2 - r_1}$$

$$\dot{Q}_{\text{sphere}} = -kA \frac{dT}{dr} = -k(4\pi r^2) \frac{C_1}{r^2} = -4\pi k C_1 = 4\pi k r_1 r_2 \frac{T_1 - T_2}{r_2 - r_1}$$

Heat Transfer Operation

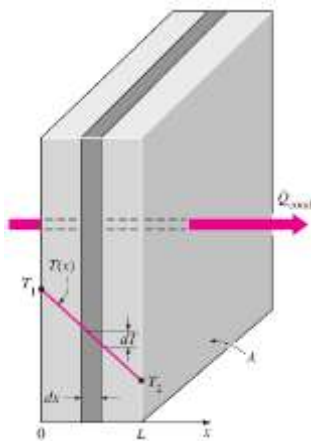
Heat generation in a solid

- Many practical heat transfer applications involve the conversion of some form of energy into thermal energy in the medium.
- Such mediums are said to involve internal heat generation, which manifests itself as a rise in temperature throughout the medium.
- Some examples of heat generation are resistance heating in wires, exothermic chemical reactions in a solid, and nuclear reactions in nuclear fuel rods where electrical, chemical, and nuclear energies are converted to heat.
- Heat generation is usually expressed per unit volume of the medium, and is denoted by $\dot{g}$, whose unit is W/m^3 .
- For example, heat generation in an electrical wire of outer radius r_0 and length L can be expressed as, where I is the electric current and R_e is the electrical resistance of the wire:

$$\dot{g} = \frac{\dot{E}_{\text{g.electric}}}{V_{\text{wire}}} = \frac{I^2 R_e}{\pi r_o^2 L} \quad (\text{W/m}^3)$$

STEADY HEAT CONDUCTION IN PLANE WALLS

Consider a plane wall of thickness L and average thermal conductivity k . The two surfaces of the wall are maintained at constant temperatures of T_1 and T_2 . For one-dimensional steady heat conduction through the wall, we have $T(x)$.



Then Fourier's law of heat conduction for the wall can be expressed as

$$\dot{Q}_{\text{cond, wall}} = -kA \frac{dT}{dx}$$

Rate of conduction heat transfer and the wall area A are constant. Thus dT/dx is constant, which temperature distribution in the wall under steady conditions is a straight line (varies linearly).

Integrating from $x = 0$ where $T(0) = T_1$, to $x = L$, where $T(L) = T_2$

Heat Transfer Operation

$$\int_{x=0}^L \dot{Q}_{\text{cond, wall}} dx = - \int_{T=T_1}^{T_2} kA dT$$

$$\dot{Q}_{\text{cond, wall}} = kA \frac{T_1 - T_2}{L}$$

The rate of heat conduction through a plane wall is proportional to the average thermal conductivity, the wall area, and the temperature difference, but is inversely proportional to the wall thickness.

Thermal Resistance

For conduction

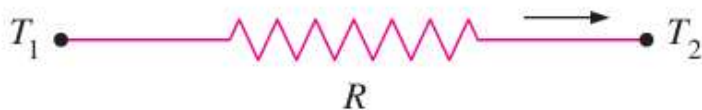
Equation for heat conduction through a plane wall can be rearranged as

$$\dot{Q}_{\text{cond, wall}} = \frac{T_1 - T_2}{R_{\text{wall}}} \quad \text{where} \quad R_{\text{wall}} = \frac{L}{kA} \quad (^\circ\text{C/W})$$

R_{wall} is the **thermal resistance of the wall against heat conduction** or simply the conduction resistance of the wall.

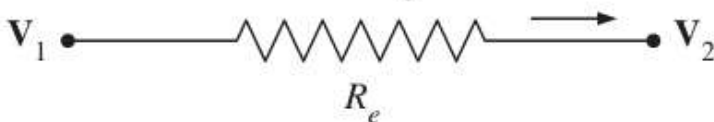
The thermal resistance of a medium depends on the geometry and the thermal properties of the medium.

$$\dot{Q} = \frac{T_1 - T_2}{R}$$



(a) Heat flow

$$I = \frac{V_1 - V_2}{R_e}$$



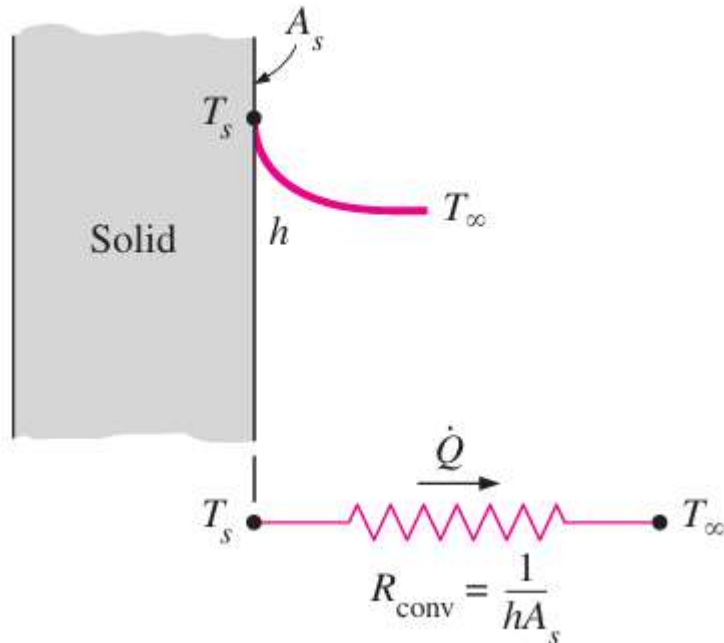
(b) Electric current flow

The equation above for heat flow is analogous to the relation for electric current flow I .

Heat Transfer Operation

For convective heat transfer

convection heat transfer from a solid surface of area A_s and temperature T_s to a fluid whose temperature sufficiently far from the surface is T_∞ , with a convection heat transfer coefficient h .



Newton's law of cooling for convection heat transfer rate can be rearranged

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty)$$

$$\dot{Q}_{\text{conv}} = \frac{T_s - T_\infty}{R_{\text{conv}}} \quad \text{where} \quad R_{\text{conv}} = \frac{1}{hA_s} \quad (^\circ\text{C/W})$$

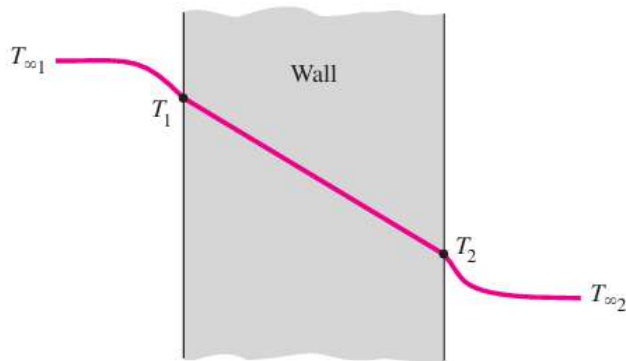
R_{conv} is the **thermal resistance of the surface against heat convection** or simply the convection resistance of the surface.

Thermal Resistance Network

Consider steady one-dimensional heat flow through a plane wall of thickness L , area A , and thermal conductivity k that is exposed to convection on both sides to fluids at temperatures T_1 and T_2 with heat transfer coefficients h_1 and h_2 .



Heat Transfer Operation



Under steady conditions,

$$\left(\begin{array}{c} \text{Rate of} \\ \text{heat convection} \\ \text{into the wall} \end{array} \right) = \left(\begin{array}{c} \text{Rate of} \\ \text{heat conduction} \\ \text{through the wall} \end{array} \right) = \left(\begin{array}{c} \text{Rate of} \\ \text{heat convection} \\ \text{from the wall} \end{array} \right)$$

$$\dot{Q} = h_1 A(T_{\infty 1} - T_1) = kA \frac{T_1 - T_2}{L} = h_2 A(T_2 - T_{\infty 2})$$

which can be rearranged as

$$\begin{aligned} \dot{Q} &= \frac{T_{\infty 1} - T_1}{1/h_1 A} = \frac{T_1 - T_2}{L/kA} = \frac{T_2 - T_{\infty 2}}{1/h_2 A} \\ &= \frac{T_{\infty 1} - T_1}{R_{\text{conv}, 1}} = \frac{T_1 - T_2}{R_{\text{wall}}} = \frac{T_2 - T_{\infty 2}}{R_{\text{conv}, 2}} \end{aligned}$$

Adding the numerators and denominators yields

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}}$$

$$R_{\text{total}} = R_{\text{conv}, 1} + R_{\text{wall}} + R_{\text{conv}, 2} = \frac{1}{h_1 A} + \frac{L}{kA} + \frac{1}{h_2 A}$$

The thermal resistances are in series, and the equivalent thermal resistance is determined by simply adding the individual resistances, just like the electrical resistances connected in series.

The rate of steady heat transfer between two surfaces is equal to the temperature difference divided by the total thermal resistance between those two surfaces.

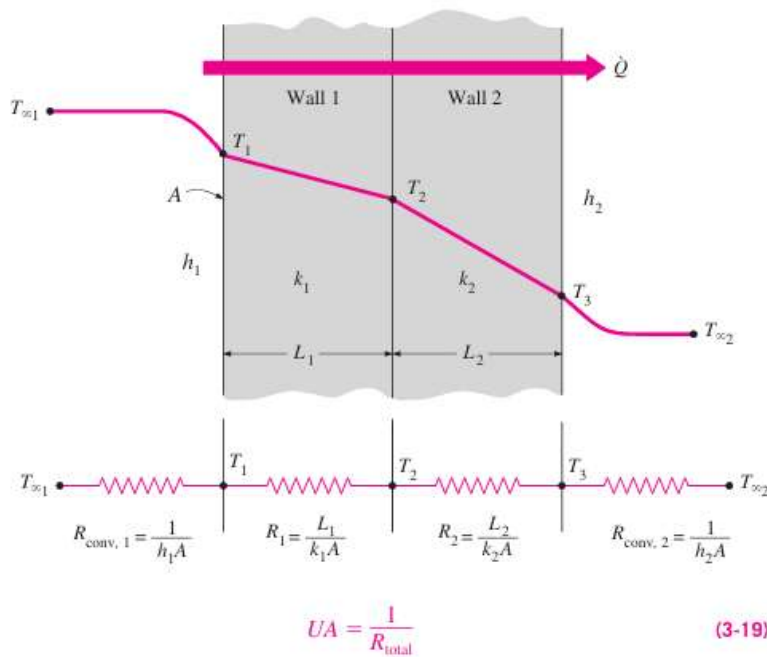
Heat Transfer Operation

The ratio of the temperature drop to the thermal resistance across any layer is constant, and thus the temperature drop across any layer is proportional to the thermal resistance of the layer. The larger the resistance, the larger the temperature drop.

$$\Delta T = \dot{Q} R$$

It is sometimes convenient to express heat transfer through a medium in an analogous manner to Newton's law of cooling as, where U is the overall heat transfer coefficient.

$$\dot{Q} = UA \Delta T$$



Therefore, for a unit area, the overall heat transfer coefficient is equal to the inverse of the total thermal resistance.

Composite Wall

In practice plane walls that consist of several layers of different materials are used.

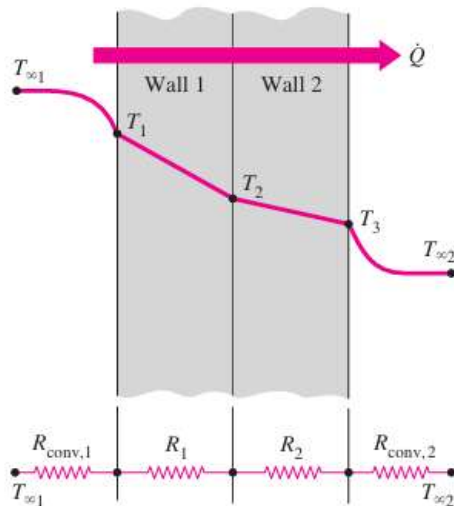
The thermal resistance concept can still be used to determine the rate of steady heat transfer through such composite walls.

Consider a plane wall that consists of two layers (such as a brick wall with a layer of insulation). The rate of steady heat transfer through this two-layer composite wall can be expressed as

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}}$$

where R_{total} is the total thermal resistance

Heat Transfer Operation



$$R_{\text{total}} = R_{\text{conv}, 1} + R_{\text{wall}, 1} + R_{\text{wall}, 2} + R_{\text{conv}, 2}$$

$$= \frac{1}{h_1 A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{1}{h_2 A}$$

From the thermal resistance network that the resistances are in series, and thus the total thermal resistance is simply the arithmetic sum of the individual thermal resistances in the path of heat flow.

Once $\dot{Q}$ is known, an unknown surface temperature T_j at any surface or interface j can be determined from

$$\dot{Q} = \frac{T_i - T_j}{R_{\text{total}, i-j}}$$

where T_i is a known temperature at location i and R_{total} , $R_{\text{total}, i-j}$ is the total thermal resistance between locations i and j .

Example: Consider a 3-m-high, 5-m-wide, and 0.3-m-thick wall whose thermal conductivity is $k = 0.9 \text{ W/m} \cdot ^\circ\text{C}$. On a certain day, the temperatures of the inner and the outer surfaces of the wall are measured to be 16°C and 2°C , respectively. Determine the rate of heat loss through the wall on that day.

Analysis Noting that the heat transfer through the wall is by conduction and the area of the wall is $A = 3 \text{ m} \times 5 \text{ m} = 15 \text{ m}^2$, the steady rate of heat transfer through the wall can be determined from Eq. 3–3 to be

$$\dot{Q} = kA \frac{T_1 - T_2}{L} = (0.9 \text{ W/m} \cdot ^\circ\text{C})(15 \text{ m}^2) \frac{(16 - 2)^\circ\text{C}}{0.3 \text{ m}} = \mathbf{630 \text{ W}}$$

Heat Transfer Operation

We could also determine the steady rate of heat transfer through the wall by making use of the thermal resistance concept from

$$\dot{Q} = \frac{\Delta T_{\text{wall}}}{R_{\text{wall}}}$$

where

$$R_{\text{wall}} = \frac{L}{kA} = \frac{0.3 \text{ m}}{(0.9 \text{ W/m} \cdot ^\circ\text{C})(15 \text{ m}^2)} = 0.02222^\circ\text{C/W}$$

Substituting, we get

$$\dot{Q} = \frac{(16 - 2)^\circ\text{C}}{0.02222^\circ\text{C/W}} = 630 \text{ W}$$

Example: Consider a 0.8-m-high and 1.5-m-wide double-pane window consisting of two 4-mm-thick layers of glass ($k = 0.78 \text{ W/m} \cdot ^\circ\text{C}$) separated by a 10-mm-wide stagnant air space ($k = 0.026 \text{ W/m} \cdot ^\circ\text{C}$). Determine the steady rate of heat transfer through this double-pane window and the temperature of its inner surface for a day during which the room is maintained at 20°C while the temperature of the outdoors is 10°C . Take the convection heat transfer coefficients on the inner and outer surfaces of the window to be $h_1 = 10 \text{ W/m}^2 \cdot ^\circ\text{C}$ and $h_2 = 40 \text{ W/m}^2 \cdot ^\circ\text{C}$, which includes the effects of radiation.

area of the window is again $A = 0.8 \text{ m} \times 1.5 \text{ m} = 1.2 \text{ m}^2$, the individual resistances are evaluated from their definitions to be

$$R_i = R_{\text{conv}, 1} = \frac{1}{h_1 A} = \frac{1}{(10 \text{ W/m}^2 \cdot ^\circ\text{C})(1.2 \text{ m}^2)} = 0.08333^\circ\text{C/W}$$

$$R_1 = R_3 = R_{\text{glass}} = \frac{L_1}{k_1 A} = \frac{0.004 \text{ m}}{(0.78 \text{ W/m} \cdot ^\circ\text{C})(1.2 \text{ m}^2)} = 0.00427^\circ\text{C/W}$$

$$R_2 = R_{\text{air}} = \frac{L_2}{k_2 A} = \frac{0.01 \text{ m}}{(0.026 \text{ W/m} \cdot ^\circ\text{C})(1.2 \text{ m}^2)} = 0.3205^\circ\text{C/W}$$

$$R_o = R_{\text{conv}, 2} = \frac{1}{h_2 A} = \frac{1}{(40 \text{ W/m}^2 \cdot ^\circ\text{C})(1.2 \text{ m}^2)} = 0.02083^\circ\text{C/W}$$

Noting that all three resistances are in series, the total resistance is

$$\begin{aligned} R_{\text{total}} &= R_{\text{conv}, 1} + R_{\text{glass}, 1} + R_{\text{air}} + R_{\text{glass}, 2} + R_{\text{conv}, 2} \\ &= 0.08333 + 0.00427 + 0.3205 + 0.00427 + 0.02083 \\ &= 0.4332^\circ\text{C/W} \end{aligned}$$

Then the steady rate of heat transfer through the window becomes

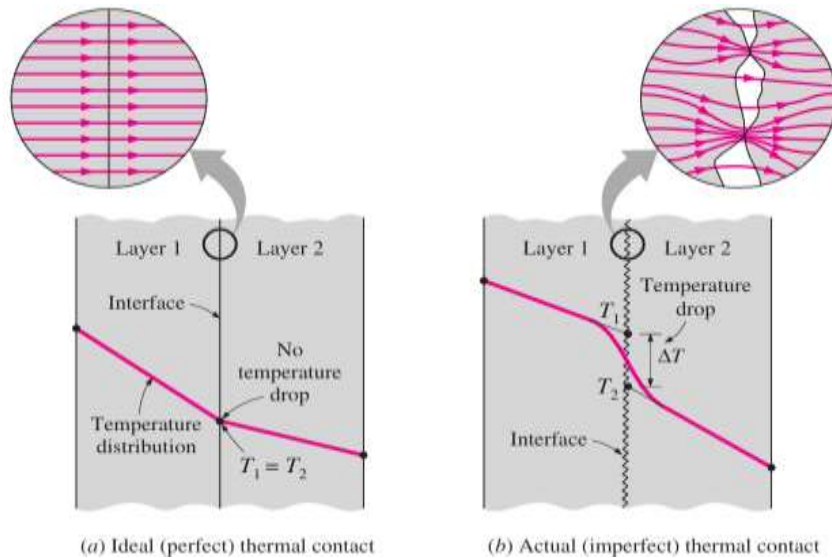
$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}} = \frac{[20 - (-10)]^\circ\text{C}}{0.4332^\circ\text{C/W}} = \mathbf{69.2 \text{ W}}$$

Heat Transfer Operation

The inner surface temperature of the window in this case will be

$$T_1 = T_{\infty 1} - \dot{Q} R_{\text{conv}, 1} = 20^\circ\text{C} - (69.2 \text{ W})(0.08333^\circ\text{C/W}) = \mathbf{14.2^\circ\text{C}}$$

THERMAL CONTACT RESISTANCE



- In the analysis of heat conduction through multilayer solids, we assumed “perfect contact” at the interface of two layers, and thus no temperature drop at the interface.
- This would be the case when the surfaces are perfectly smooth and they produce a perfect contact at each point.
- In reality, however, even flat surfaces that appear smooth to the eye turn out to be rather rough when examined under a microscope, with numerous peaks and valleys.
- That is, a surface is microscopically rough no matter how smooth it appears to be.
- When two such surfaces are pressed against each other, the peaks will form good material contact but the valleys will form voids filled with air.
- As a result, an interface will contain numerous air gaps of varying sizes that act as insulation because of the low thermal conductivity of air.
- Thus, an interface offers some resistance to heat transfer, and this resistance per unit interface Cold fluid area is called the thermal contact resistance, R_c .

Consider heat transfer through two metal rods of cross-sectional area A that are pressed against each other.

Heat transfer through the interface of these two rods is the sum of the heat transfers through the solid contact spots and the gaps in the noncontact areas and can be expressed as

$$\dot{Q} = \dot{Q}_{\text{contact}} + \dot{Q}_{\text{gap}}$$

It can also be expressed in an analogous manner to Newton’s law of cooling as

$$\dot{Q} = h_c A \Delta T_{\text{interface}}$$

Heat Transfer Operation

where A is the apparent interface area (which is the same as the cross-sectional area of the rods) and $T_{\text{interface}}$ is the effective temperature difference at the interface.

The quantity h_c , which corresponds to the convection heat transfer coefficient, is called the thermal contact conductance

$$h_c = \frac{\dot{Q}/A}{\Delta T_{\text{interface}}}$$

It is related to thermal contact resistance by

$$R_c = \frac{1}{h_c} = \frac{\Delta T_{\text{interface}}}{\dot{Q}/A}$$

R_c represents thermal contact resistance per unit area.

The thermal resistance for the entire interface is obtained by dividing R_c by the apparent interface area A .

Thermal contact resistance is observed to decrease with decreasing surface roughness and increasing interface pressure.

$$R_{c, \text{insulation}} = \frac{L}{k}$$

Example: The thermal contact conductance at the interface of two 1-cm-thick aluminum plates is measured to be $11,000 \text{ W/m}^2 \cdot ^\circ\text{C}$. Determine the thickness of the aluminum plate whose thermal resistance is equal to the thermal resistance of the interface between the plates.

SOLUTION The thickness of the aluminum plate whose thermal resistance is equal to the thermal contact resistance is to be determined.

Properties The thermal conductivity of aluminum at room temperature is $k = 237 \text{ W/m} \cdot ^\circ\text{C}$ (Table A-3).

Analysis Noting that thermal contact resistance is the inverse of thermal contact conductance, the thermal contact resistance is

$$R_c = \frac{1}{h_c} = \frac{1}{11,000 \text{ W/m}^2 \cdot ^\circ\text{C}} = 0.909 \times 10^{-4} \text{ m}^2 \cdot ^\circ\text{C/W}$$

For a unit surface area, the thermal resistance of a flat plate is defined as

$$R = \frac{L}{k}$$

where L is the thickness of the plate and k is the thermal conductivity. Setting $R = R_c$, the equivalent thickness is determined from the relation above to be

$$L = kR_c = (237 \text{ W/m} \cdot ^\circ\text{C})(0.909 \times 10^{-4} \text{ m}^2 \cdot ^\circ\text{C/W}) = 0.0215 \text{ m} = \mathbf{2.15 \text{ cm}}$$

Heat Transfer Operation

HEAT CONDUCTION IN CYLINDERS AND SPHERES

Consider a **long cylindrical layer** (such as a circular pipe) of inner radius r_1 , outer radius r_2 , length L , and average thermal conductivity k .

Fourier's law of heat conduction for heat transfer through the cylindrical layer can be expressed as

$$\dot{Q}_{\text{cond, cyl}} = -kA \frac{dT}{dr}$$

integrating from

$$r = r_1, \text{ where } T(r_1) = T_1, \text{ to } r = r_2, \text{ where } T(r_2) = T_2$$

$$\int_{r=r_1}^{r_2} \frac{\dot{Q}_{\text{cond, cyl}}}{A} dr = - \int_{T=T_1}^{T_2} k dT$$

$$A = 2\pi rL$$

$$\dot{Q}_{\text{cond, cyl}} = 2\pi Lk \frac{T_1 - T_2}{\ln(r_2/r_1)}$$

$$\dot{Q}_{\text{cond, cyl}} = \frac{T_1 - T_2}{R_{\text{cyl}}}$$

$$R_{\text{cyl}} = \frac{\ln(r_2/r_1)}{2\pi Lk} = \frac{\ln(\text{Outer radius/Inner radius})}{2\pi \times (\text{Length}) \times (\text{Thermal conductivity})}$$

is the thermal resistance of the cylindrical layer against heat conduction, or simply the conduction resistance of the cylinder layer.

For a spherical layer

$$\int_{r=r_1}^{r_2} \frac{\dot{Q}_{\text{cond, cyl}}}{A} dr = - \int_{T=T_1}^{T_2} k dT$$

$$A = 4\pi r^2$$

$$\dot{Q}_{\text{cond, sph}} = \frac{T_1 - T_2}{R_{\text{sph}}}$$

$$R_{\text{sph}} = \frac{r_2 - r_1}{4\pi r_1 r_2 k} = \frac{\text{Outer radius} - \text{Inner radius}}{4\pi(\text{Outer radius})(\text{Inner radius})(\text{Thermal conductivity})}$$

is the thermal resistance of the spherical layer against heat conduction, or simply the conduction resistance of the spherical layer.

Heat Transfer Operation

Multilayered Cylinders and Spheres

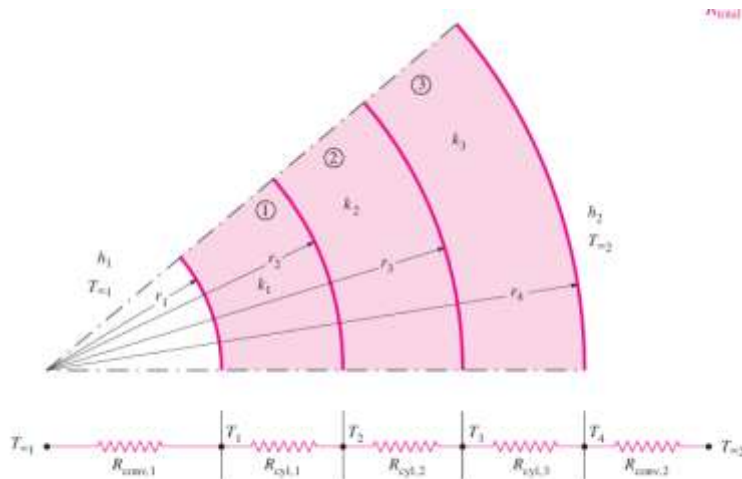


FIGURE 3-26
The thermal resistance network for heat transfer through a three-layered composite cylinder subjected to convection on both sides.

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}}$$

where R_{total} is the total thermal resistance

$$\begin{aligned} R_{\text{total}} &= R_{\text{conv}, 1} + R_{\text{cyl}, 1} + R_{\text{cyl}, 2} + R_{\text{cyl}, 3} + R_{\text{conv}, 2} \\ &= \frac{1}{h_1 A_1} + \frac{\ln(r_2/r_1)}{2\pi L k_1} + \frac{\ln(r_3/r_2)}{2\pi L k_2} + \frac{\ln(r_4/r_3)}{2\pi L k_3} + \frac{1}{h_2 A_4} \end{aligned}$$

where $A_1 = 2\pi r_1 L$ and $A_4 = 2\pi r_4 L$.

Example: Steam at $T_1 = 320^\circ\text{C}$ flows in a cast iron pipe ($k = 80 \text{ W/m}\cdot^\circ\text{C}$) whose inner and outer diameters are $D_1 = 5 \text{ cm}$ and $D_2 = 5.5 \text{ cm}$, respectively. The pipe is covered with 3-cm-thick glass wool insulation with $k = 0.05 \text{ W/m}\cdot^\circ\text{C}$. Heat is lost to the surroundings at $T_2 = 5^\circ\text{C}$ by natural convection and radiation, with a combined heat transfer coefficient of $h_2 = 18 \text{ W/m}^2\cdot^\circ\text{C}$. Taking the heat transfer coefficient inside the pipe to be $h_1 = 60 \text{ W/m}^2\cdot^\circ\text{C}$, determine the rate of heat loss from the steam per unit length of the pipe. Also determine the temperature drops across the pipe shell and the insulation.

tances in series and is given in Fig. 3–29. Taking $L = 1 \text{ m}$, the areas of the surfaces exposed to convection are determined to be

$$\begin{aligned} A_1 &= 2\pi r_1 L = 2\pi(0.025 \text{ m})(1 \text{ m}) = 0.157 \text{ m}^2 \\ A_3 &= 2\pi r_3 L = 2\pi(0.0575 \text{ m})(1 \text{ m}) = 0.361 \text{ m}^2 \end{aligned}$$

Heat Transfer Operation

Then the individual thermal resistances become

$$R_i = R_{\text{conv}, 1} = \frac{1}{h_1 A} = \frac{1}{(60 \text{ W/m}^2 \cdot ^\circ\text{C})(0.157 \text{ m}^2)} = 0.106^\circ\text{C/W}$$

$$R_1 = R_{\text{pipe}} = \frac{\ln(r_2/r_1)}{2\pi k_1 L} = \frac{\ln(2.75/2.5)}{2\pi(80 \text{ W/m} \cdot ^\circ\text{C})(1 \text{ m})} = 0.0002^\circ\text{C/W}$$

$$R_2 = R_{\text{insulation}} = \frac{\ln(r_3/r_2)}{2\pi k_2 L} = \frac{\ln(5.75/2.75)}{2\pi(0.05 \text{ W/m} \cdot ^\circ\text{C})(1 \text{ m})} = 2.35^\circ\text{C/W}$$

$$R_o = R_{\text{conv}, 2} = \frac{1}{h_2 A_3} = \frac{1}{(18 \text{ W/m}^2 \cdot ^\circ\text{C})(0.361 \text{ m}^2)} = 0.154^\circ\text{C/W}$$

Noting that all resistances are in series, the total resistance is determined to be

$$R_{\text{total}} = R_i + R_1 + R_2 + R_o = 0.106 + 0.0002 + 2.35 + 0.154 = 2.61^\circ\text{C/W}$$

Then the steady rate of heat loss from the steam becomes

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}} = \frac{(320 - 5)^\circ\text{C}}{2.61^\circ\text{C/W}} = \mathbf{121 \text{ W}} \quad (\text{per m pipe length})$$

The heat loss for a given pipe length can be determined by multiplying the above quantity by the pipe length L .

The temperature drops across the pipe and the insulation are determined from Eq. 3-17 to be

$$\begin{aligned}\Delta T_{\text{pipe}} &= \dot{Q} R_{\text{pipe}} = (121 \text{ W})(0.0002^\circ\text{C/W}) = \mathbf{0.02^\circ\text{C}} \\ \Delta T_{\text{insulation}} &= \dot{Q} R_{\text{insulation}} = (121 \text{ W})(2.35^\circ\text{C/W}) = \mathbf{284^\circ\text{C}}\end{aligned}$$

That is, the temperatures between the inner and the outer surfaces of the pipe differ by 0.02°C , whereas the temperatures between the inner and the outer surfaces of the insulation differ by 284°C .

CRITICAL RADIUS OF INSULATION

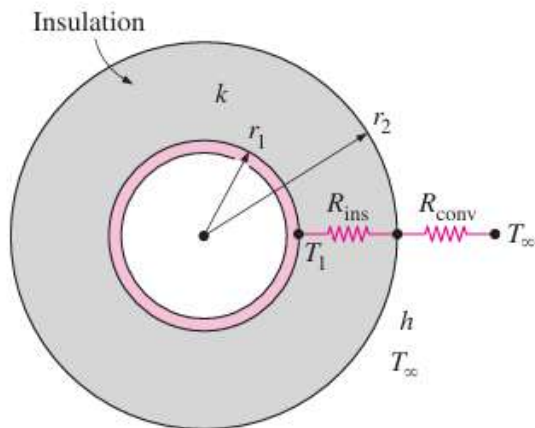
The thicker the insulation, the lower the heat transfer rate. This is expected, since the heat transfer area A is constant, and adding insulation always increases the thermal resistance of the wall without increasing the convection resistance.

Adding insulation to a cylindrical pipe or a spherical shell, however, is a different matter. The additional insulation increases the conduction resistance of the insulation layer but decreases the convection resistance of the surface because of the increase in the outer surface area for convection.

The heat transfer from the pipe may increase or decrease, depending on which effect dominates. Consider a cylindrical pipe of outer radius r_1 whose outer surface temperature T_1 is maintained constant.

Heat Transfer Operation

The pipe is now insulated with a material whose thermal conductivity is k and outer radius is r_2 .



Heat is lost from the pipe to the surrounding medium at temperature T_∞ , with a convection heat transfer coefficient h .

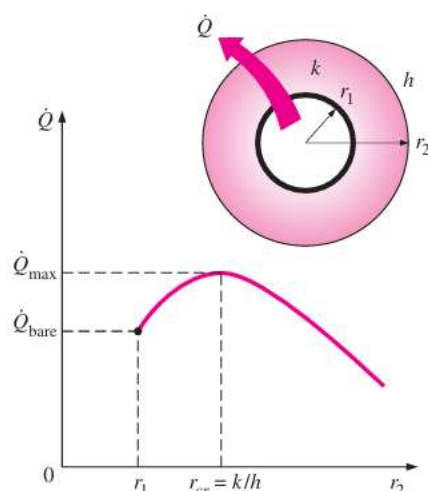
The rate of heat transfer from the insulated pipe to the surrounding air can be expressed as

$$\dot{Q} = \frac{T_1 - T_\infty}{R_{ins} + R_{conv}} = \frac{T_1 - T_\infty}{\frac{\ln(r_2/r_1)}{2\pi Lk} + \frac{1}{h(2\pi r_2 L)}}$$

The variation of $\dot{Q}$ with the outer radius of the insulation r_2 is plotted.

The value of r_2 at which $\dot{Q}$ reaches a maximum is determined from the requirement that $d\dot{Q}/dr_2 = 0$ (zero slope).

Performing the differentiation and solving for r_2 yields the critical radius of insulation for a cylindrical body to be



$$r_{cr, cylinder} = \frac{k}{h}$$

Heat Transfer Operation

The critical radius of insulation depends on the thermal conductivity of the insulation k and the external convection heat transfer coefficient h . The rate of heat transfer from the cylinder increases with the addition of insulation for $r_2 < r_{cr}$, reaches a maximum when $r_2 = r_{cr}$, and starts to decrease for $r_2 > r_{cr}$. Thus, insulating the pipe may actually increase the rate of heat transfer from the pipe instead of decreasing it when $r_2 < r_{cr}$.

The critical radius of insulation for a spherical shell is

$$r_{cr, sphere} = \frac{2k}{h}$$

Where k is the thermal conductivity of the insulation and h is the convection heat transfer coefficient on the outer surface.

Example: A 3-mm-diameter and 5-m-long electric wire is tightly wrapped with a 2-mm thick plastic cover whose thermal conductivity is $k = 0.15 \text{ W/m} \cdot ^\circ\text{C}$. Electrical measurements indicate that a current of 10 A passes through the wire and there is a voltage drop of 8 V along the wire. If the insulated wire is exposed to a medium at $T = 30^\circ\text{C}$ with a heat transfer coefficient of $h = 12 \text{ W/m}^2 \cdot ^\circ\text{C}$, determine the temperature at the interface of the wire and the plastic cover in steady operation. Also determine whether doubling the thickness of the plastic cover will increase or decrease this interface temperature.

Analysis Heat is generated in the wire and its temperature rises as a result of resistance heating. We assume heat is generated uniformly throughout the wire and is transferred to the surrounding medium in the radial direction. In steady operation, the rate of heat transfer becomes equal to the heat generated within the wire, which is determined to be

$$\dot{Q} = \dot{W}_e = VI = (8 \text{ V})(10 \text{ A}) = 80 \text{ W}$$

The thermal resistance network for this problem involves a conduction resistance for the plastic cover and a convection resistance for the outer surface in series, as shown in Fig. 3–32. The values of these two resistances are determined to be

$$A_2 = (2\pi r_2)L = 2\pi(0.0035 \text{ m})(5 \text{ m}) = 0.110 \text{ m}^2$$

$$R_{\text{conv}} = \frac{1}{hA_2} = \frac{1}{(12 \text{ W/m}^2 \cdot ^\circ\text{C})(0.110 \text{ m}^2)} = 0.76^\circ\text{C/W}$$

$$R_{\text{plastic}} = \frac{\ln(r_2/r_1)}{2\pi kL} = \frac{\ln(3.5/1.5)}{2\pi(0.15 \text{ W/m} \cdot ^\circ\text{C})(5 \text{ m})} = 0.18^\circ\text{C/W}$$

and therefore

$$R_{\text{total}} = R_{\text{plastic}} + R_{\text{conv}} = 0.76 + 0.18 = 0.94^\circ\text{C/W}$$

Notes created by Dr. Panchami Patel

Reference Books:

- Cengel A. Yunus. "Heat Transfer – A Practical Approach", McGraw Hill.

Heat Transfer Operation

Then the interface temperature can be determined from

$$\dot{Q} = \frac{T_1 - T_\infty}{R_{\text{total}}} \quad \longrightarrow \quad T_1 = T_\infty + \dot{Q}R_{\text{total}} \\ = 30^\circ\text{C} + (80 \text{ W})(0.94^\circ\text{C/W}) = \mathbf{105^\circ\text{C}}$$

$$r_{\text{cr}} = \frac{k}{h} = \frac{0.15 \text{ W/m} \cdot ^\circ\text{C}}{12 \text{ W/m}^2 \cdot ^\circ\text{C}} = 0.0125 \text{ m} = 12.5 \text{ mm}$$

which is larger than the radius of the plastic cover. Therefore, increasing the thickness of the plastic cover will *enhance* heat transfer until the outer radius of the cover reaches 12.5 mm. As a result, the rate of heat transfer $\dot{Q}$ will *increase* when the interface temperature T_1 is held constant, or T_1 will *decrease* when $\dot{Q}$ is held constant, which is the case here.